Massimiliano Iaschi

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EDUCATION

Georgia Institute of Technology, Atlanta, GA — *Expected May 2026*
B.S. in Mechanical Engineering — GPA: 3.97/4.00 (Major: 4.00/4.00)
Minors: Robotics, Bio-inspired Design

Liceo Scientifico Statale Stanislao Cannizzaro, Rome, Italy — *September 2016 – Jul 2021*
Diploma di Esame di Stato — GPA: 9.85/10 (Final Exam: 100/100)

RELEVANT EXPERIENCE

Robotic Systems Lab (RSL), Visiting Researcher, ETH Zurich, Switzerland — Aug – Dec 2025
Advisor: Prof. Marco Hutter, Prof. Baxi Chong; Supervisor: Robert Baines (postdoc).
Research objective: enabling lizard-like shape-morphing robots that adapt their morphology to the terrain.

- Co-design of lizard-like robots to optimize locomotion across different terrains: using robotics simulation and learning to understand how different morphological parameters interact with each other, with different control policies, and with different environments.

CRAB Lab, Bio-Inspired Robotics Undergraduate Researcher, Georgia Tech, USA — Aug 2023 – Present
Advisor: Prof. Daniel I. Goldman; Supervisor: Baxi Chong (postdoc), Juntao He (PhD student).
Research objective: leveraging mechanical intelligence to enhance multi-legged robotic locomotion performance in complex environments.

- Focus on mechatronics and biologically inspired design, control systems design, physics-based modeling, and SLAM integration.
- Characterized Scuttle, a novel multi-segmented, multi-C-legged robotic platform: investigated its dynamics, contact interactions, and terrain-dependent locomotion physics on flat, granular, and complex substrates; developed several open-loop controllers; and co-led a journal manuscript currently in preparation with PhD student Jianfeng Lin.
- Conceived and led the design of a fully novel, multi-segmented robot developed from scratch, featuring a cable-driven compliant joint enabling both a rotational DoF in the sagittal plane and a prismatic DoF for coupling vertical and peristaltic traveling waves; designed the robot's open-loop controller, formulated the scientific hypotheses, and led both the robotics and biological experiments as well as the manuscript preparation.
- Contributed to the mechatronic design and prototyping of a novel centipede-inspired robot featuring foot sensors and bio-inspired antennae; assisted in developing the feedforward controller that enabled successful climbing behaviors for obstacles up to 5 times the robot's height through tactile sensing, performing lab and field experiments, and editing the manuscript.
- Designed, prototyped, and tested a hexapod robot used to develop a novel contact-planning framework via spin models duality; foot and controller design to enable stair-climbing in a low-torque multi-legged robotic platform; explored and prototyped soft-robotic manipulators and bionic antenna designs; designed and prototyped various experimental setups for high-speed cameras to conduct experiments on real centipedes in various conditions, conducted experiments, and analyzed biological experiment data.

Autonomous Systems Lab (ASL), Visiting Researcher, ETH Zurich, Switzerland — May – Aug 2023
Advisor: Prof. Roland Siegwart; Supervisor: multiple.

- Tethys Project, worked on the Proteus underwater robot for exploration and rescue: autonomously tested the robot in the wild, designed a legged retractable automatic locking mechanism to ease robot deployment, contributed to the design of state-estimation algorithms, high-level design of a soft-robotic tail, took two Georgia Tech courses, and attended the ETH ROS course for Master Students.

PoWeR Lab, Bio-Mechatronics Undergraduate Researcher, Georgia Tech, USA — Jan 2022 – Apr 2023
Advisor: Prof. Gregory Sawicky; Supervisor: Amro Alshareef (PhD student).

- Contributed to the recreation in the lab of Stanford's open-source *Bump'Em* platform (designed and implemented its PID control system in Simulink) for studying human balance and gait; contributed CAD modeling and rapid prototyping across multiple bio-mechatronic projects.

Siena Robotics & Systems Lab, Summer Robotics Intern, Univ. of Siena, Italy Jun – Jul 2022
Advisor: Prof. Domenico Prattichizzo; Supervisor: multiple.

- Validated the hypothesis that thermal modulation at the fingertip can create the illusion of a broader contact surface for robotic haptic applications: designed and assembled a simple custom hardware setup for generating thermal illusions, designed a feedforward temperature control system in LabVIEW to enable real-time modulation of haptic perception, took two Georgia Tech courses.

PUBLICATIONS

1. **Iaschi, M.**, Chong, B., Wang, T., Lin, J., He, J. (2025). *Addition of a Peristaltic Wave Improves Multi-Legged Locomotion Performance on Complex Terrains.* IEEE Int. Conf. on Robotics and Automation (ICRA), Atlanta.
2. He, J., Chong, B., **Iaschi, M.**, Nienhusser, V., Goldman, D. I. (2025). *Tactile Sensing Enables Vertical Obstacle Negotiation for Elongate Many-Legged Robots.* Robotics: Science and Systems (RSS).
3. Teder, E., Chong, B., He, J., Wang, T., **Iaschi, M.**, Soto, D., Goldman, D. I. (2025). *Effective Self-Righting Strategies for Elongate Multi-Legged Robots.* IEEE ICRA.
4. Soto, D., Erickson, E., Pierce, C., Diaz, K., **Iaschi, M.**, Lee, A., Goldman, D. I. (2024). *Legged Locomotion in Lattices: Centipede Traversal of Obstacle-Rich Environments. Under review, Annals of the New York Academy of Sciences.*

CONFERENCE TALKS

Iaschi, M. "Addition of a Peristaltic Wave Improves Multi-Legged Locomotion" — presented at ICRA (Atlanta), APS (Anaheim), SICB (Atlanta), 2025.
Iaschi, M. "Centipede Locomotion on Complex Environments" — iPoLS Annual Meeting, Trieste, Italy, Jun 2024.

ROBOTICS PEER REVIEWER

Reviewer for one bipedal-locomotion-related manuscript (ICRA 2025) and selected reviewer for one soft-robotics manuscript (RoboSoft 2025).

ADDITIONAL SKILLS

Technical: Python, C++/ROS, MATLAB, PyTorch, Isaac Sim/Lab, Mujoco, Optuna, SolidWorks, Git, Linux, mechatronic prototyping (3D printing, laser cutting, water jet cutting, soldering), Arduino, Simulink, LabVIEW, LaTeX.
Selected Coursework:
Robotics & Mechatronics: Autonomous Mobile Robots, Automation and Robotics, Fundamentals of Mechatronics (Spring 2026), Creative Decisions and Design;
Modeling & Controls: Modeling and Control of Motion Systems (Spring 2026), System Dynamics, Dynamics of Rigid Bodies, Differential Equations, Statistics and Applications;
Bio-Inspired Design: Bio-Inspired Design, Behavioral Biology.
Languages: English and Italian (fluent); Spanish (B1); French (A2).
Creative: Author of adventure novels for young adults (Italian), Renaissance art enthusiast.